Self assessment:

1. What Works

The project successfully implements an end-to-end semantic speech command classification pipeline using the all-MiniLM-L6-v2 Sentence Transformer and a Cosine Similarity Classifier. The system supports 14 predefined semantic commands and accurately classifies semantically similar inputs without relying on exact keyword matching. It also incorporates threshold-based Out-of-Scope (OOS) rejection to identify unsupported commands. The model was successfully exported to ONNX and optimized using INT8 dynamic quantization, reducing the model size and improving inference speed while maintaining comparable classification performance. Evaluation on both clean and noisy datasets demonstrated the effectiveness and robustness of the proposed approach.

2. What Does Not Work

The current implementation supports only a predefined set of commands and accepts text input rather than direct speech input. The cosine similarity threshold is manually selected and may require adjustment for different datasets or applications. Additionally, the system has been evaluated only on a desktop CPU and not on an actual edge device.

3. Possible Improvements

Future improvements include integrating a speech-to-text module for real voice input, automatically optimizing the similarity threshold, precomputing training embeddings to reduce startup time, and expanding the command dataset to support additional semantic commands. Evaluating alternative lightweight embedding models could also improve performance.

4. Deployment Considerations

To deploy this system as a real-world product, the quantized ONNX model should be integrated with an application using ONNX Runtime Mobile. A speech recognition module would be required to convert spoken commands into text before classification. Additional testing on target edge devices, along with memory, latency, and battery consumption analysis, would be necessary to ensure reliable real-time performance.